Cryptanalysis of the Sharafi–Daghigh digital signature scheme

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Abstract

This paper is devoted to the study of the Ring-LWE-based digital signature scheme proposed by Sharafi and Daghigh, especially, the cryptanalysis of this scheme. The Sharafi and Daghigh's scheme is inspired by the Lindner–Peikert encryption paradigm and adopts a hash-and-sign approach via the Fiat–Shamir transformation. The security claims rely on the assumed hardness of the Ring-LWE and Ring-SIS problems, whose definitions and properties are well studied in last two decades. We demonstrate that this scheme is not secure and generalise our analysis to the analogous scheme in the plain-LWE setting.

Keywords: LWE, SIS, Digital Signature, Cryptanalysis, Quantum Safe

1 Introduction

The impending threat of quantum computing [1–3] has made it necessary to move from traditional to post-quantum cryptographic methods. Among the various possible new families, lattice-based cryptography has attracted much attention because of its strong worst-case hardness promises and flexibility in algorithmic use cases.

In this field, the Learning With Errors (LWE) problem [4] and its ring-based extension, Ring-LWE [5, 6], are primitive hardness assumptions that form the foundation for numerous contemporary constructions. Lattice-based cryptography has shown a promising alternative to classical public-key constructions with further applications, like Fully-Homomorphic Encryption (FHE), constructing privacy-preserving systems and much more. NIST has standardized two lattice-based constructions, namely Kyber [7] and Dilithium [8], their standardized version are ML-KEM [9] and [10] respectively.

In this paper, we focus on the detailed analysis and breakdown of a Ring-LWE-based digital signature scheme, proposed by Sharafi and Daghigh [11], inspired from the Lindner- Peikert cryptosystem. The scheme is in the hash-and-sign paradigm with the Fiat-Shamir transformation, providing efficiency and compactness. With the use of structured lattices via cyclotomic number fields, it is key and signature size-reducing while it is conjectured of being quantum-resistant in the Quantum Random Oracle Model (QROM). The main contribution of this paper has been to introduce a flaw-by-design attack that compromises unforgeability of the scheme in chosen-message settings. Rather than depending on inconsistent choice of parameters, the noise distributions, or simply Fiat-Shamir transform, this attack takes advantage of a more profound issue: the signing algorithm lacks enough algebraic binds to the secret key. As a result, an adversary having access only to a public key can easily compute valid signatures for any input message–thus compromising even the most fundamental ideals of public-key-only attack security, and consequently, also that of the UF-CMA notion.

We also show that this same vulnerability persists in the plain setting, therefore the same attack can be adapted to module variant, the generalization of the scheme, where the base ring is replaced by R_q^k . The attack carries over component-wise, meaning that this vulnerability is not a coincidence but a deep-seated part of the fundamental design paradigm. Since the underlying signature scheme has a vulnerability to forgery when either its private key or an oracle is missing, no remedial actions are suggested. The purpose of this paper is to clarify and systematically record this intrinsic weakness through theory as well as through experimental evidence.

We begin by providing the mathematical and algorithmic prerequisites to understand the scheme and our proposed attack in section 2. We then restate the signature scheme proposed by Sharafi and Daghigh in section 3, its correctness and verification, and deconstruct its security reduction in section 4. A proof of concept implementation of the scheme and our attack is given in the section 5. Lastly, a conclusion is given in section 7.

2 Preliminaries

In this section we revisit some lattice basics and lattice hard problems that most of lattice-based cryptography relies on.

Definition 1 A lattice $\mathcal{L}$ formally is a discrete subgroup of $\mathbb{R}^n$ for a norm $\|\cdot\|$. A lattice can be represented by a basis $B = \{b_1, \ldots, b_m\}$, for $b_i \in \mathbb{R}^n$.

It is known that, for every lattice $\mathcal{L}$, there is a non-zero vector in $\mathcal{L}$ of minimal euclidean norm, and finding such a vector is the well-known problem of the shortest vector problem.

Definition 2 Shortest vector problem (SVP) The shortest vector problem is **given**: $B \in \mathbb{Z}^{n \times m}$, **to find**: $z \in \mathbb{Z}^m \setminus \{0\}$, such that $\|Bz\|$ has minimal length.

Notably, lattice bases are not unique; any unimodular transformation preserves the lattice structure [5]. The quality of a lattice basis it measure on how near orthogonal and shorter the basis vectors are.

The process of transforming an initial lattice basis into one with improved geometric properties, termed lattice reduction [12, 13], employs various algorithmic strategies that trade computational efficiency for output quality. For instance, the LLL algorithm [12] achieves polynomial-time complexity but yields vectors with exponential approximation factors relative to the shortest vector. In contrast, HKZ reduction guarantees a basis containing the shortest non-zero vector, albeit at the cost of exponential runtime. A detailed comparison of these methods and their trade-offs can be found in the survey [14] and the recent preprint survey [15]. Notably, advancements in lattice reduction remain an active research area, with recent breakthroughs such as the Progressive BKZ framework [16] and highly optimized implementations like the General Sieve Kernel (G6K) [17] pushing the boundaries of practical efficiency.

For more details and discussion of the hardness of the above problems, we refer the reader to [4, 14].

2.1 Lattice-Based Cryptography

Lattice-based cryptography has emerged as a cornerstone of post-quantum security due to its resilience against quantum algorithms. The foundational work by Ajtai [18] established a critical link between average-case and worst-case hardness of lattice problems, enabling cryptographic constructions based on robust computational assumptions.

2.2 Learning With Errors (LWE)

The Learning With Errors (LWE) problem, introduced by Regev [4], underpins many lattice-based schemes. Let χ be an error distribution over $\mathbb{Z}_q$. For a secret $s \in \mathbb{Z}_q^n$, the LWE distribution $\mathcal{A}_{s,\chi}$ generates samples $(a, b) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, where $b = \langle s, a \rangle + e \mod q$ with $e \leftarrow \chi$. Two variants exist:

- **Search-LWE**: Recover s given polynomially many samples.
- **Decision-LWE**: Distinguish LWE samples from uniform ones.

LWE's security relies on the conjectured intractability of solving noisy linear equations.

2.3 Ring-LWE

To improve efficiency, Lyubashevsky et al. [6] proposed Ring-LWE, an algebraic variant operating over polynomial rings. Let $R = \mathbb{Z}[X]/\langle f(X)\rangle$ for a cyclotomic polynomial $f(X) = X^n + 1$ (typically n a power of 2) and $R_q = R/qR$. For a secret $s \in R_q$, a Ring-LWE sample $(a, b) \in R_q \times R_q$ satisfies $b = s \cdot a + e \mod q$, where $e \leftarrow \chi$. The decision variant requires distinguishing such samples from uniform pairs. Cyclotomic rings are preferred due to their algebraic structure and proven hardness guarantees [6].

2.4 Short Integer Solution (SIS) and Ring-SIS

The SIS problem, introduced by Ajtai [18], involves finding a short non-zero vector $z \in \mathbb{Z}^m$ - i.e. for a small $\beta > 0$ find $z \in \mathbb{Z}^m$ such that $\|z\| \leq \beta$ - such that $A\mathbf{z} = 0 \mod q$ for a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$. Its ring analogue, Ring-SIS [5, 6], operates over R_q: given $\mathbf{a} = (a_1, \ldots, a_m) \in R_q^m$, find $z \in R^m$ with $\|z\| \leq \beta$ satisfying $\sum_{i=1}^m a_i z_i = 0 \mod q$ for a small $\beta > 0$. Both problems are reducible to worst-case lattice problems, forming the basis for collision-resistant hash functions and digital signatures.

3 Review of the Proposed Digital Signature Scheme

Most of the content of this section is from the original paper of Sharif and Daghigh, we recall the most related part to our attack from their paper, we refer the reader to their original paper for further explanations [11]. The signature scheme under study is built upon the Ring-LWE assumption and inspired by the Lindner–Peikert encryption system. It follows a "hash-and-sign" paradigm using the Fiat–Shamir transformation and operates over a cyclotomic ring $R = \mathbb{Z}[X]/(X^n+1)$ where n is a power of two. Let $R_q = R/qR$ for a modulus q. The encoding and decoding mechanisms play a critical role in the correctness and security of the scheme.

3.1 Digital signature syntax and security notion (UF-CMA)

A *digital signature scheme* is a triple of probabilistic polynomial-time algorithms $(\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Verify})$ where $\mathsf{KeyGen}(1^\lambda)$ outputs a key pair (pk, sk), $\mathsf{Sign}(sk, m)$ outputs a signature σ, and $\mathsf{Verify}(pk, m, \sigma) \in \{\mathsf{accept}, \mathsf{reject}\}$. Correctness requires that for all messages m, $\Pr[\mathsf{Verify}(pk, m, \mathsf{Sign}(sk, m)) = \mathsf{accept}] = 1 - o(1)$. Unforgeability under chosen-message attack (UF-CMA) is defined by the standard game in which an adversary given pk and access to a signing oracle must not output a valid signature on a fresh message with non-negligible probability.

3.2 Encoding and decoding: domain, range, and tolerance

Let Σ denote the message space (bitstrings of appropriate length in the original proposal). The scheme uses deterministic maps

$$\text{encode} : \Sigma \to R_q, \qquad \text{decode} : R_q \to \Sigma,$$

where $R_q = R/qR$. They satisfy an error-tolerance property: there exists a threshold t such that for all $m \in \Sigma$ and all errors e with $\|e\|_\infty \leq t$,

$$\text{decode}(\text{encode}(m) + e) = m.$$

In [11], the decoding threshold is set to $t = \lfloor q/4 \rfloor$.

3.3 Key Generation

The signer samples two secret elements $r_1, r_2 \leftarrow \chi_k$ from a discrete Gaussian distribution χ_k over R_q, and a public element $a \in R_q$ is sampled uniformly at random. The public key is computed as:

$$p = r_1 - ar_2 \in R_q.$$

Algorithm 1 Key Generation

1: **Input:** Ring R_q, error distribution χ_k
2: Sample $r_1, r_2 \leftarrow \chi_k$
3: Sample $a \leftarrow R_q$ uniformly
4: Compute $p = r_1 - ar_2$
5: **Output:** Public key (a, p), Secret key r_2

3.4 Signing procedure

Given a message $m \in \{0,1\}^n$, the signer hashes it and encodes it as $\bar{m} = \text{encode}(H(m)) \in R_q$, using an injective encoding function. The signer samples ephemeral noise terms $e_i \leftarrow \chi_e$ for $i = 1, 2, 3, 4$ and computes:

$$C_1 = pe_1 + e_2, C_2 \quad = pae_1 + e_3 + \bar{m}, C_3 = ae_2 + e_4$$

Then, the signer computes:

$$\omega = \text{decode}(ar_2e_1), \quad h = H(m \,\|\, H(\omega)).$$

The signature is the tuple (C_1, C_2, C_3, h).

Algorithm 2 Signing

1: **Input:** Message m, secret key r_2, public parameters (a, p)
2: Sample $e_1, e_2, e_3, e_4 \leftarrow \chi_e$
3: Compute $\bar{m} \leftarrow \text{encode}(H(m))$
4: Compute:

$$C_1 = pe_1 + e_2, C_2 = pae_1 + e_3 + \bar{m}, C_3 = ae_2 + e_4$$

5: Compute $\omega \leftarrow \text{decode}(ar_2e_1)$
6: Compute $h \leftarrow H(m \,\|\, H(\omega))$
7: **Output:** Signature (C_1, C_2, C_3, h)

3.5 Verification procedure

Given a message m and a signature (C_1, C_2, C_3, h), the verifier computes:

$$\bar{m} = \text{encode}(H(m)),$$

and checks the following two conditions:

1. $\text{decode}(C_2 - aC_1 + C_3) = \bar{m}$,
2. $h \stackrel{?}{=} H(m \,\|\, H(\text{decode}(-C_1)))$.

Algorithm 3 Verification

1: **Input:** Message m, signature (C_1, C_2, C_3, h), public key (a, p)
2: Compute $\bar{m} \leftarrow \text{encode}(H(m))$
3: Compute $\omega' \leftarrow \text{decode}(-C_1)$
4: Compute $h' \leftarrow H(m \,\|\, H(\omega'))$
5: Check:

$$\textbf{(i)} \quad \text{decode}(C_2 - aC_1 + C_3) \stackrel{?}{=} \bar{m}$$

$$\textbf{(ii)} \quad h' \stackrel{?}{=} h$$

6: **Accept** if both conditions hold; otherwise **Reject**

Remark 1 (On $H(m \parallel H(\omega))$ vs $H(m \parallel \omega)$) The original construction computes the Fiat–Shamir challenge as $h \leftarrow H(m \parallel H(\omega))$. Using $H(\omega)$ provides a fixed-length, canonical input to the hash function and avoids serialisation ambiguities. This choice does *not* affect our cryptanalysis: in a forgery, the adversary can compute $\omega' = \text{decode}(-C_1)$ from its own C_1 and hence can compute $H(m \parallel H(\omega'))$ just as the verifier does.

3.6 Design Choices: Encoding, Error Distribution, and Noise Thresholds

The encode/decode functions are designed such that:

$$\text{decode}(\text{encode}(m) + e) = m$$

for all $m \in \{0,1\}^n$ and "small enough" noise e in R_q. The error terms are sampled from discrete Gaussian distributions χ_k (for secrets) and χ_e (for ephemeral randomness) centered at zero. The success of decoding (i.e., correctness of verification) is guaranteed with high probability provided the noise remains below a threshold $t = \lfloor \frac{q}{4} \rfloor$.

These choices aim to balance correctness and security. However, as we demonstrate in the next section, the tight coupling of noise-dependent correctness and Fiat–Shamir commitments creates a critical structural vulnerability.

4 Cryptanalysis: A Flaw-by-Design Attack

4.1 Description of the Attack

The attack we present is not based on solving any underlying hard lattice problem such as Ring-LWE or Ring-SIS. Instead, it is a structural attack that exploits the algebraic relationships within the signature generation and verification process. It allows an adversary to generate valid signatures on arbitrary messages without querying the signer, thus violating unforgeability in a catastrophic way.

Our attack replicates the structure of a valid signature without access to the secret key. The adversary proceeds as follows:

1. Choose an arbitrary message $m \in \{0,1\}^n$ and define $\bar{m} = \text{encode}(H(m))$.
2. Sample $e_1, e_2, e_3, e_4 \leftarrow \chi_e$ from the same small-error distribution used by the signer.
3. Compute the following components:

$$C_1 = pe_1 + e_2, C_2 = pae_1 + e_3 + \bar{m}, C_3 = ae_2 + e_4.$$

4. Instead of compute ω as the signer, the adversary computes directly $\omega = \text{decode}(-C_1)$.
5. compute $h = H(m \,\|\, H(\omega))$.

The forged signature (C_1, C_2, C_3, h) passes verification, because:

$$\begin{aligned} C_2 - aC_1 + C_3 &= (pae_1 + e_3 + \bar{m}) - a(pe_1 + e_2) + ae_2 + e_4 \\ &= (pae_1 - ape_1) + e_3 + \bar{m} + e_4 = \bar{m} + (e_3 + e_4). \end{aligned}$$

Since $e_3 + e_4$ is small with high probability, the decoding succeeds:

$$\text{decode}(C_2 - aC_1 + C_3) = \bar{m}.$$

Thus, the signature is accepted by the verifier, even though the message was never signed by the legitimate party.

4.2 Proof of Insecurity

We now formally prove that the scheme fails to satisfy the unforgeability under chosen-message attacks (UF-CMA) requirement. Specifically, we show that a probabilistic polynomial-time adversary can generate a valid signature on an arbitrary message without querying the signing oracle.

Theorem 1 *The Ring-LWE-based signature scheme of Sharafi and Daghigh is insecure in the UF-CMA model. There exists a polynomial-time adversary that can forge signatures on arbitrary messages with overwhelming probability.*

Proof Let $\mathcal{A}$ be an adversary who does not make any signing queries. On input the public key (a, p), $\mathcal{A}$ a message $m \in \{0,1\}^n$ to be signed. $\mathcal{A}$ computes $\bar{m} = \text{encode}(H(m))$. Then it

samples $e_1, e_2, e_3, e_4 \leftarrow \chi_e$, and computes:

$$C_1 = pe_1 + e_2, C_2 = pae_1 + e_3 + \bar{m}, C_3 = ae_2 + e_4.$$

$\mathcal{A}$ checks whether $\text{decode}(C_2 - aC_1 + C_3)$ equals m or not, if not, $\mathcal{A}$ repeats from step 2. This loop always finish with high probability due to the fact $e_3 + e_4$ is small with high probability and hence the value $C_2 - aC_1 + C_3$ is decoded to m. Now the adversary computes $\omega = \text{decode}(-C_1)$. Finally, $\mathcal{A}$ outputs the signature (C_1, C_2, C_3, h) where $h = H(m \,\|\, H(\omega))$. This signature passes verification because:

$$\text{decode}(C_2 - aC_1 + C_3) = \text{decode}(\bar{m} + e_3 + e_4) = m,$$

with high probability, provided that the sum $e_3 + e_4$ lies within the decoding radius. Since all elements involved are small by design, this condition is met with overwhelming probability. The second condition is valid by constructions since $\omega = \text{decode}(-C_1)$, and therefore the verifier will find the recompute h' equals to h in the given signature. As conclusion, the adversary succeeds in forging a signature on a new message without given any valid signature, nor the private key to sign, which violates the UF-CMA security definition. □

5 Experimental Proof of Concept

To confirm the viability of the attack in practice, we implemented a prototype of the signature scheme and the flaw-by-design forgery algorithm (Algorithm 4) in SageMath. The primary goal was to demonstrate that for any chosen message, it is possible to generate a valid signature that passes verification without access to the private key.

Algorithm 4 Flaw-by-Design Forgery Attack

Require: Public key (a, p), m the message to be signed, decoding function decode, error distribution χ_e

Ensure: Forged signature (C_1, C_2, C_3, h) on a message $m \in \{0,1\}^n$

1: Compute $\bar{m} \leftarrow \text{encode}(H(m))$
2: Sample $e_1, e_2, e_3, e_4 \leftarrow \chi_e$
3: Compute $C_1 \leftarrow pe_1 + e_2$
4: Compute $C_2 \leftarrow pae_1 + e_3 + \bar{m}$
5: Compute $C_3 \leftarrow ae_2 + e_4$
6: Compute $h \leftarrow H(m \,\|\, H(\text{decode}(-C_1)))$
7: **return** (C_1, C_2, C_3, h)

Experimental platform

All experiments were conducted on a single machine with the following specifications:

CPU	Intel Core i7-class processor
RAM	16 GB
OS	Ubuntu 22.04 LTS (64-bit)
SageMath	9.6
Python	3.10

Reported runtimes correspond to average wall-clock time over the stated trials.

Setup

We used the parameter set recommended in the original proposal:

- Ring dimension: $n = 256$
- Modulus: $q = 7681$
- Error distribution: centered discrete Gaussian χ_e with standard deviation $\sigma = 2.5$
- Decoding threshold: $t = \lfloor \frac{q}{4} \rfloor = 1920$

We generated public keys using the original scheme key generation algorithm, but all signatures were produced independently using the attack using only the public key and a random message as inputs.

Procedure

For each of 1000 randomly selected messages:

1. We computed $\bar{m} = \text{encode}(H(m))$.
2. Sampled noise vectors $e_1, e_2, e_3, e_4 \leftarrow \chi_e$.
3. Computed (C_1, C_2, C_3) using the public key and the formulas of the attack.
4. Checked whether $\text{decode}(C_2 - aC_1 + C_3) = m$.

Code availability

A fully reproducible proof-of-concept implementation of the attack is provided as supplementary material in the notebook: https://github.com/ndrahmani/attack_sharafi_daghigh.

Results

Our experiments show that the forged signature passes the decoding check with a high success rate . In rare cases where the decode fails, re-sampling the error terms until the procedure succeed. The entire forgery process requires less than 10ms per message on a standard laptop.

These results confirm that the attack is not merely theoretical but also practical and efficient. It does not rely on any oracle access, nor does it attempt to recover the secret key. Instead, it forges valid signatures by exploiting the deterministic algebraic structure of the scheme.

Interpretation

The experimental findings corroborate the theoretical proof of insecurity: the attacker can produce an arbitrary number of valid signatures on any messages of their choosing. Since the verification condition only relies on smallness of the decoding error, and the error is attacker-controlled, the correctness mechanism becomes a source of forgery rather than security.

This proves that the signature scheme is not only mathematically broken but also practically able to forge in real-world settings.

6 Extended Cryptanalysis to the plain and Module Variants

The structural vulnerability extends naturally to the plain-LWE generalization of the scheme. In the module setting, the ring R_q is replaced by R_q^k, but the encode/decode mechanism and the correctness-based verification condition remain essentially unchanged.

The attack scales component-wise: since each coordinate of the module vector behaves like an independent instance of the base scheme, an adversary can apply the same reconstruction logic to each coordinate. In fact, the increased redundancy in higher-dimensional module structures may make the attack even easier to mount.

Therefore, the paradigm regardless of whether it is instantiated over Ring-LWE, Module-LWE or even plain-LWE remains fundamentally broken.

7 Conclusion

We have demonstrated key-only attack from a structural flaw in the design of a Ring-LWE-based digital signature scheme that invalidates its unforgeability, even under standard hardness assumptions and quantum models. Our attack exploits the fact that an attacker does not need to know the signing keys to generate valid signatures, as the generated error terms can be done also by the attacker knowing the verification key only, since this error terms are not bounded to the secret signing key, then the are easy to be generated.

The flaw generalizes directly to the plain-LWE setting, suggesting that this design paradigm is fundamentally unsound for digital signatures. As the attack targets the protocol structure itself rather than its parametrization, no patch or countermeasure is suggested. This work reinforces the need to re-evaluate design principles when adapting encryption primitives to signatures, particularly in the post-quantum setting.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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